

FINANCIAL ECONOMETRICS

Exam

2 hours. Closed book exam.

The problem and exercise are independent.

Problem : It is often impractical to use all of the assets that are traded in a market. One alternative is to find a portfolio composed of a given set of M stocks that tracks the market portfolio most closely in the sense of minimizing the variance of the difference in returns. Such portfolios are called "tracking-efficient portfolios".

A "portfolio $(a_0, \mathbf{a})$ " means a portfolio of composition a_0 in the risk-free asset and $\mathbf{a} = (a_1, \dots, a_M)'$ in the M risky assets. We set $\boldsymbol{\omega}_{\mathbf{a}} = (a_1 p_{1,t}, \dots, a_M p_{M,t})'$ where $p_{i,t}$ denotes the price of asset i at time t ($i = 1, \dots, M$). Let $y_{i,t+1} = \frac{p_{i,t+1}}{p_{i,t}}$ and $\mathbf{y}_{t+1} = (y_{1,t+1}, \dots, y_{M,t+1})'$. By convention the risk-free asset has price 1 at time t and return r_t between time t and $t + 1$. On the same period, the market portfolio has return $y_{t+1}^{(m)}$.

Suppose that the joint distribution of the risky assets and the market portfolio has the following conditional moments (given the available information at time t) :

$$E_t \begin{pmatrix} \mathbf{y}_{t+1} \\ y_{t+1}^{(m)} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\mu}_t \\ \mu_t^{(m)} \end{pmatrix}, \quad \text{Var}_t \begin{pmatrix} \mathbf{y}_{t+1} \\ y_{t+1}^{(m)} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\Omega}_t & \boldsymbol{\Lambda}_t \\ \boldsymbol{\Lambda}_t' & \sigma_{t,m}^2 \end{pmatrix}$$

where $\boldsymbol{\mu}_t \in \mathbb{R}^M$ and $\boldsymbol{\Omega}_t$ is a positive-definite $M \times M$ matrix. Let also $\tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t - (1 + r_t)\mathbf{e}$ where $\mathbf{e} = (1, \dots, 1)' \in \mathbb{R}^M$.

Denote by V_t the value of a portfolio composed, at time t , of a_0 units of the risk-free asset and a_i units of assets i , for $i = 1, \dots, M$:

$$V_t = a_0 + \boldsymbol{\omega}_{\mathbf{a}}' \mathbf{e}.$$

We wish to find the portfolio of initial value $V_t = v > 0$ which minimizes the conditional variance of the difference $V_{t+1}/V_t - y_{t+1}^{(m)}$, under the constraint that the expected net return of the portfolio has an assigned level s_t (with $s_t \geq 1 + r_t$) :

$$\begin{cases} \min_{a_0, \mathbf{a}} \text{Var}_t \left(\frac{V_{t+1}}{V_t} - y_{t+1}^{(m)} \right) \\ \text{under the constraint } E_t \left(\frac{V_{t+1}}{V_t} \right) = s_t \end{cases}$$

1. After elimination of a_0 write the program to be solved in terms of $\boldsymbol{\omega}_{\mathbf{a}}$.

2. Show that the solutions (i.e. the compositions in risky assets of the tracking-efficient portfolios) have the form

$$\omega_{\mathbf{a}} = v\Omega_t^{-1}(\Lambda_t + \lambda_t\tilde{\mu}_t),$$

where λ_t is a "constant" depending (in particular) on s_t .

3. In which case do the classes of "tracking-efficient portfolios" and "efficient portfolios" coincide?
4. Is there an analogue of the two fund separation theorem? If so, give an expression for two funds that can be used to obtain every tracking-efficient portfolio.
5. In the mean-variance diagram, we define a "tracking-efficient frontier" of all tracking-efficient portfolios. Give the form of the tracking-efficient frontier and compare it with the efficient frontier.

Exercise : Suppose that the net excess returns of M risky assets satisfy the regression

$$z_{i,t} = \alpha_i + \beta_i F_t + \beta_i^{(m)} z_t^{(m)} + \epsilon_{i,t}, \quad i = 1, \dots, M,$$

where F_t is a common factor, $F_t \in \mathbb{R}$, $z_t^{(m)}$ is the net excess return of the market portfolio, $\epsilon_{i,t}$ are error terms, and $\alpha = (\alpha_1, \dots, \alpha_M)'$, $\beta = (\beta_1, \dots, \beta_M)'$, and $\beta^{(m)} = (\beta_1^{(m)}, \dots, \beta_M^{(m)})'$ are vector of parameters. A set of n observations of the variables $z_{i,t}$, F_t and $z_t^{(m)}$ is available.

1. Write the model in vector form and explain why (without giving a proof) the system can be estimated "equation-by-equation" under appropriate assumptions on the error terms.
2. Let an estimator $(\hat{\alpha}_n, \hat{\beta}_n, \hat{\beta}_n^{(m)})$ of $(\alpha, \beta, \beta^{(m)})$ and assume that the following asymptotic distribution holds :

$$\sqrt{n} \begin{pmatrix} \hat{\alpha}_n - \alpha \\ \hat{\beta}_n - \beta \\ \hat{\beta}_n^{(m)} - \beta^{(m)} \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(0, \Gamma := \begin{pmatrix} \Gamma_{11} & \Gamma_{12} & \Gamma_{13} \\ \Gamma_{21} & \Gamma_{22} & \Gamma_{23} \\ \Gamma_{31} & \Gamma_{32} & \Gamma_{33} \end{pmatrix} \right),$$

where Γ is positive definite, and the Γ_{ij} are $M \times M$ matrices for which estimators consistent $\hat{\Gamma}_{ij}$ can be constructed.

- (a) Write the null assumption corresponding to the CAPM and propose a test. How would you interpret a rejection of the CAPM?
- (b) Does the model allow to test the Fama-French model?
- (c) In order to increase the adequacy of the model, one may wish to introduce dynamic betas instead of a static regression. How can this be done? briefly explain the possible ways of extending the model while keeping the econometric tractability.

SOLUTION

(NB : all questions are not corrected in detail)

Problem :

1. We have $V_t = v = a_0 + \omega'_a \mathbf{e}$, $V_{t+1} = a_0(1 + r_t) + \omega'_a \mathbf{y}_{t+1}$. Hence

$$E_t \left(\frac{V_{t+1}}{V_t} \right) = \frac{1}{v} \{a_0(1 + r_t) + \omega'_a \mu_t\} = (1 + r_t) + \frac{1}{v} \omega'_a \tilde{\mu}_t$$

and

$$\text{Var}_t \left(\frac{V_{t+1}}{V_t} - y_{t+1}^{(m)} \right) = \frac{1}{v^2} \omega'_a \Omega_t \omega_a + \sigma_{t,m}^2 - \frac{2}{v} \omega'_a \Lambda_t.$$

The program can be written

$$\begin{cases} \min_{\omega_a} \omega'_a \Omega_t \omega_a - 2v \omega'_a \Lambda_t \\ u.c. \quad (1 + r_t) + \frac{1}{v} \omega'_a \tilde{\mu}_t = s_t \end{cases}$$

2. Introducing the Lagrangian, the program writes

$$\begin{cases} \min_{\omega_a, \lambda_t} \mathcal{L}(\omega_a, \lambda_t) \\ \mathcal{L}(\omega_a, \lambda_t) = \omega'_a \Omega_t \omega_a - 2v \omega'_a \Lambda_t - 2\lambda_t v^2 ((1 + r_t) + \frac{1}{v} \omega'_a \tilde{\mu}_t - s_t). \end{cases}$$

The first-order conditions are

$$\begin{cases} \frac{\partial \mathcal{L}(\omega_a, \lambda_t)}{\partial \omega_a} = 2(\Omega_t \omega_a - v \Lambda_t - v \lambda_t \tilde{\mu}_t) = 0 \\ \frac{\partial \mathcal{L}(\omega_a, \lambda_t)}{\partial \lambda_t} = -2v^2 ((1 + r_t) + \frac{1}{v} \omega'_a \tilde{\mu}_t - s_t) = 0. \end{cases}$$

It follows that $\omega_a = v \Omega_t^{-1} (\Lambda_t + \lambda_t \tilde{\mu}_t)$ where

$$\lambda_t = \frac{s_t - (1 + r_t) - \Lambda'_t \Omega_t^{-1} \tilde{\mu}_t}{\tilde{\mu}'_t \Omega_t^{-1} \tilde{\mu}_t}.$$

3. Efficient portfolios are such that ω_a is proportional to $\Omega_t^{-1} \tilde{\mu}_t$. Tracking-efficient portfolios are thus efficient in the usual sense if and only if the vectors Λ_t and $\tilde{\mu}_t$ are colinear.
4. The composition is the risk-free asset of a tracking-efficient portfolio is

$$a_0 = v - \omega'_a \mathbf{e} = v \{1 - (\Lambda_t + \lambda_t \tilde{\mu}_t)' \Omega_t^{-1} \mathbf{e}\}.$$

It follows that every efficient tracking portfolio can be written as

$$\begin{pmatrix} a_0 \\ \mathbf{a} \end{pmatrix} = v \begin{pmatrix} 1 - \mathbf{\Lambda}'_t \mathbf{\Omega}_t^{-1} \mathbf{e} \\ \{\text{diag}(\mathbf{p}_t)\}^{-1} \mathbf{\Omega}_t^{-1} \mathbf{\Lambda}_t \end{pmatrix} + \lambda_t v \begin{pmatrix} -\tilde{\boldsymbol{\mu}}'_t \mathbf{\Omega}_t^{-1} \mathbf{e} \\ \{\text{diag}(\mathbf{p}_t)\}^{-1} \mathbf{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t \end{pmatrix} \\ := \mathbf{P}_0 + \lambda_t \mathbf{P}^*,$$

where $\mathbf{P}_0$ does not depend on the risk aversion of the investor, while $\mathbf{P}^*$ has value 0 at time t (short positions balancing long positions).

5. The conditional mean and variance of a tracking-efficient portfolio are, respectively,

$$\begin{aligned} m_t &= E_t(V_{t+1}) = v(1 + r_t + \mathbf{\Lambda}'_t \mathbf{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t + \lambda_t \tilde{\boldsymbol{\mu}}'_t \mathbf{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t) \\ \sigma_t^2 &= \text{Var}_t(V_{t+1}) = v^2(\lambda_t^2 \tilde{\boldsymbol{\mu}}'_t \mathbf{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t + 2\lambda_t \tilde{\boldsymbol{\mu}}'_t \mathbf{\Omega}_t^{-1} \mathbf{\Lambda}_t + \mathbf{\Lambda}'_t \mathbf{\Omega}_t^{-1} \mathbf{\Lambda}_t). \end{aligned}$$

Elimination of λ_t yields

$$\sigma_t^2 = \frac{\{m_t - (1 + r_t)v\}^2}{\tilde{\boldsymbol{\mu}}'_t \mathbf{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t} + v^2 c_t$$

where

$$c_t = \frac{(\mathbf{\Lambda}'_t \mathbf{\Omega}_t^{-1} \mathbf{\Lambda}_t)(\tilde{\boldsymbol{\mu}}'_t \mathbf{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t) - (\tilde{\boldsymbol{\mu}}'_t \mathbf{\Omega}_t^{-1} \mathbf{\Lambda}_t)^2}{\tilde{\boldsymbol{\mu}}'_t \mathbf{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t} \geq 0$$

by the Cauchy-Schwarz inequality.

In the mean-variance diagram, the tracking-efficient frontier is a semiparabola with vertex $(\sigma_t^2 = c_t v^2, m_t = (1 + r_t)v)$, obtained by an horizontal translation of the efficient frontier.

Exercise :

1. Letting $\mathbf{z}_t = (z_{1,t}, \dots, z_{M,t})'$ and $\boldsymbol{\epsilon}_t = (\epsilon_{1,t}, \dots, \epsilon_{M,t})'$, the model writes

$$\mathbf{z}_t = \boldsymbol{\alpha} + \boldsymbol{\beta} F_t + \boldsymbol{\beta}^{(m)} z_t^{(m)} + \boldsymbol{\epsilon}_t.$$

The model can be estimated by GLS or, equivalently, by OLS equation-by-equation since the regressors, F_t and $z_t^{(m)}$ are common to the M assets. One has to assume that the errors satisfy : $E(\boldsymbol{\epsilon}_t | F_t, z_t^{(m)}) = \mathbf{0}$, $V(\boldsymbol{\epsilon}_t | F_t, z_t^{(m)}) = \boldsymbol{\Sigma}$, where $\boldsymbol{\Sigma}$ is a positive-definite matrix.

2. (a) The null assumption writes $H_0 : \boldsymbol{\alpha} = \boldsymbol{\beta} = \mathbf{0}$, that is $\mathbf{R}\boldsymbol{\theta} = \mathbf{0}$ where $\mathbf{R} = \begin{pmatrix} \mathbf{I}_N & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_N & \mathbf{0} \end{pmatrix}$. A Wald test statistic is

$$T_n = n(\hat{\boldsymbol{\alpha}}'_n, \hat{\boldsymbol{\beta}}'_n) \begin{pmatrix} \hat{\mathbf{\Gamma}}_{11} & \hat{\mathbf{\Gamma}}_{12} \\ \hat{\mathbf{\Gamma}}_{21} & \hat{\mathbf{\Gamma}}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \hat{\boldsymbol{\alpha}}_n \\ \hat{\boldsymbol{\beta}}_n \end{pmatrix}, \quad \text{with } T_n \xrightarrow{d} \chi^2(2M) \quad \text{under } H_0.$$

A critical region for H_0 at level $\alpha \in (0, 1)$ thus has the form $\{T_n \geq \chi^2_{1-\alpha}(2M)\}$. Rejection of the CAPM can be due to a violation of the hypotheses (see the course) or to the use of a poor proxy of the market portfolio (for instance a financial index).

- (b) The Fama-French model requires 3 specific factors, which is not permitted here.
- (c) See Chapter 5 of the course.

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The exercises are independent.

Notations reminder : Consider an economy with N risky assets and one risk-free asset. A "portfolio $P_{a_0, \mathbf{a}}$ " means a portfolio of composition a_0 in the risk-free asset and $\mathbf{a} = (a_1, \dots, a_N)'$ in the N risky assets. We set $\boldsymbol{\omega}_{\mathbf{a}} = (a_1 p_{1,t}, \dots, a_N p_{N,t})'$ where $p_{i,t}$ denotes the price of asset i at time t ($i = 1, \dots, N$). Let $y_{i,t+1} = \frac{p_{i,t+1}}{p_{i,t}}$ and $\mathbf{y}_{t+1} = (y_{1,t+1}, \dots, y_{N,t+1})'$, the vector of gross returns of the risky assets. By convention the risk-free asset has price 1 at time t and return r_t between time t and $t+1$.

The conditional moments of $\mathbf{y}_{t+1}$ (given the available information at time t) are :

$$E_t(\mathbf{y}_{t+1}) = \boldsymbol{\mu}_t, \quad \text{Var}_t(\mathbf{y}_{t+1}) = \boldsymbol{\Omega}_t$$

where $\boldsymbol{\mu}_t \in \mathbb{R}^N$ and $\boldsymbol{\Omega}_t$ is a positive-definite $N \times N$ matrix. Let also $\tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t - (1 + r_t)\mathbf{e}$ where $\mathbf{e} = (1, \dots, 1)' \in \mathbb{R}^N$.

Denote by V_t the value of a portfolio composed, at time t , of a_0 units of the risk-free asset and a_i units of assets i , for $i = 1, \dots, N$: $V_t = a_0 + \boldsymbol{\omega}'_{\mathbf{a}} \mathbf{e}$. The value of the portfolio at time $t+1$ is denoted $V_{t+1}(a_0, \mathbf{a})$.

Exercise 1 : The goal is to study Portfolio analysis based on utility functions. According to utility theory, the investor aims at maximizing the expected utility of a portfolio return, i.e. maximizing $E_t[u\{V_{t+1}(a_0, \mathbf{a})\}]$ where the utility u is an increasing real-valued function. Call $\mathcal{U}_1$ the set of increasing utility functions.

Let two portfolios $P_{a_0, \mathbf{a}}$ and $P_{b_0, \mathbf{b}}$ with the same value v at time t .

$P_{a_0, \mathbf{a}}$ is called preferable to $P_{b_0, \mathbf{b}}$ for $u \in \mathcal{U}_1$, denoted $P_{a_0, \mathbf{a}} \stackrel{u}{\succeq} P_{b_0, \mathbf{b}}$, if

$$E_t[u\{V_{t+1}(a_0, \mathbf{a})\}] \geq E_t[u\{V_{t+1}(b_0, \mathbf{b})\}],$$

provided the expectations exist.

1. $P_{a_0, \mathbf{a}}$ stochastically dominates $P_{b_0, \mathbf{b}}$, denoted $P_{a_0, \mathbf{a}} \stackrel{SD}{\succeq} P_{b_0, \mathbf{b}}$, if $P_{a_0, \mathbf{a}} \stackrel{u}{\succeq} P_{b_0, \mathbf{b}}$ for all $u \in \mathcal{U}_1$. We will prove that

$$P_{a_0, \mathbf{a}} \stackrel{SD}{\succeq} P_{b_0, \mathbf{b}} \Leftrightarrow P_t[V_{t+1}(a_0, \mathbf{a}) \geq x] \geq P_t[V_{t+1}(b_0, \mathbf{b}) \geq x], \quad \text{for all } x \in \mathbb{R}. \quad (1)$$

- (a) Prove the implication $\Rightarrow$ in (1). *Hint : take, for a given x , the utility function $u(y)$ equal to one if $y \geq x$ and to 0 otherwise.*
- (b) Prove the converse (that is $\Leftarrow$ in (1)) by computing $P_t[u(V_{t+1}(a_0, \mathbf{a})) \geq x]$ using the inverse (or generalized inverse) of u .
2. Assume that, given the available information at time t , the vector $\mathbf{y}_{t+1}$ follows a Gaussian distribution.
 - (a) Determine the conditional distribution of $\boldsymbol{\omega}'_{\mathbf{a}} \mathbf{y}_{t+1}$. Write $V_{t+1}(a_0, \mathbf{a})$ in function of v and $\boldsymbol{\omega}_{\mathbf{a}}$, and discuss the existence of portfolios which stochastically dominate the other portfolios.
 - (b) Consider the utility $u(x) = -\exp(-\gamma x)$ where $\gamma > 0$. Find the solutions of the optimization program

$$\max_{a_0, \mathbf{a}} E_t[u\{V_{t+1}(a_0, \mathbf{a})\}]$$

for portfolios of initial value $V_t(a_0, \mathbf{a}) = v$. Compare with the usual mean-variance analysis and interpret γ . *Hint : if $X \sim \mathcal{N}(0, 1)$, we have $E[e^{-\gamma X}] = e^{\gamma^2/2}$.*

Exercise 2 Consider the factor model

$$\mathbf{y}_{t+1} = \mathbf{c} + \mathbf{B}\mathbf{f}_{t+1} + \mathbf{u}_{t+1}, \quad E_t(\mathbf{u}_{t+1} \mid \mathbf{f}_{t+1}) = \mathbf{0}, \quad E_t(\mathbf{u}_{t+1}\mathbf{u}'_{t+1} \mid \mathbf{f}_{t+1}) = \boldsymbol{\Sigma}$$

where $\mathbf{f}_{t+1} \in \mathbb{R}^K$, $\mathbf{B} = (B_{i,j})$ is a $N \times K$ matrix of factor loadings, $\mathbf{c} \in \mathbb{R}^N$. We assume that $E_t(\mathbf{f}_{t+1}) = \mathbf{\Lambda}$ and that $\text{Var}_t(\mathbf{f}_{t+1}) = \boldsymbol{\Gamma}$ is a positive-definite matrix.

1. Show that the return of a portfolio with composition $\mathbf{a} = (a_1, \dots, a_N)'$ in risky assets has the form $\bar{\boldsymbol{\omega}}'_{\mathbf{a}} \mathbf{y}_{t+1}$, for some vector $\bar{\boldsymbol{\omega}}_{\mathbf{a}}$.
2. Compute $E_t(\bar{\boldsymbol{\omega}}'_{\mathbf{a}} \mathbf{y}_{t+1})$ and $V_t(\bar{\boldsymbol{\omega}}'_{\mathbf{a}} \mathbf{y}_{t+1})$. Explain the decomposition in systemic and idiosyncratic risks in the latter formula.
3. We say that the idiosyncratic risk is diversifiable if $\lim_{N \rightarrow \infty} \bar{\boldsymbol{\omega}}'_{\mathbf{a}} \bar{\boldsymbol{\omega}}_{\mathbf{a}} = 0$ implies that $\lim_{N \rightarrow \infty} \bar{\boldsymbol{\omega}}'_{\mathbf{a}} \boldsymbol{\Sigma} \bar{\boldsymbol{\omega}}_{\mathbf{a}} = 0$. A portfolio such that the latter convergence holds is called diversifiable.
 - (a) Suppose that $\boldsymbol{\Sigma}$ is diagonal, $\boldsymbol{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_N^2)$. Give a simple condition on the σ_i^2 for diversification to hold. Deduce that the equally weighted portfolio such that $\bar{\boldsymbol{\omega}}'_{\mathbf{a}} \mathbf{e} = 1$ is diversifiable under this condition.
 - (b) Show that diversification holds if the spectral radius $\rho(\boldsymbol{\Sigma})$ of $\boldsymbol{\Sigma}$ is bounded when N increases ($\rho(\boldsymbol{\Sigma}) = \max_{1, \dots, N} |\lambda_i|$ where the λ_i 's are the eigenvalues of $\boldsymbol{\Sigma}$).
 - (c) Suppose that there is some ordering in the components of $\mathbf{u}_{t+1}$ such that

$$\text{Cov}_t(u_{i,t+1}, u_{j,t+1}) = \rho^{|i-j|}, \quad \text{for some } \rho \in (0, 1), \text{ and } i, j \in \{1, \dots, N\}.$$

Show that the equally weighted portfolio such that $\bar{\boldsymbol{\omega}}'_{\mathbf{a}} \mathbf{e} = 1$ is diversifiable.

- (d) For two diversifiable portfolios with compositions $\mathbf{a}$ and $\mathbf{b}$ find an approximation of $\text{Cov}_t(\bar{\boldsymbol{\omega}}'_{\mathbf{a}} \mathbf{y}_{t+1}, \bar{\boldsymbol{\omega}}'_{\mathbf{b}} \mathbf{y}_{t+1})$ for large N . Note that the two portfolios can be highly correlated even if their compositions are orthogonal, in the sense that $\bar{\boldsymbol{\omega}}'_{\mathbf{a}} \bar{\boldsymbol{\omega}}_{\mathbf{b}} = 0$.

SOLUTION

(NB : Not all questions are corrected in details)

Exercise 1 :

1. (a) Let $u(y) = 1_{y \geq x}$. We have $u \in \mathcal{U}_1$. Moreover

$$E_t[u\{V_{t+1}(a_0, \mathbf{a})\}] = P_t[V_{t+1}(a_0, \mathbf{a}) \geq x] \geq E_t[u\{V_{t+1}(b_0, \mathbf{b})\}].$$

which proves the implication.

- (b) Suppose that the right-hand side of the implication holds. Then, assuming u is strictly increasing and continuous, we have, for all x ,

$$\begin{aligned} P_t[u\{V_{t+1}(a_0, \mathbf{a})\} \geq x] &= P_t[V_{t+1}(a_0, \mathbf{a}) \geq u^{-1}(x)] \\ &\geq P_t[V_{t+1}(b_0, \mathbf{b}) \geq u^{-1}(x)] = P_t[u\{V_{t+1}(b_0, \mathbf{b})\} \geq x]. \end{aligned}$$

In the general case, one can use the generalized inverse u^- of u . If u is right-continuous, we have $u(x) \geq y \Leftrightarrow x \geq u^-(y)$ so the same argument applies. This implies that the equivalence holds if $\mathcal{U}_1$ is restricted to the set of increasing and right-continuous functions.

2. (a) The variable $\omega'_a \mathbf{y}_{t+1}$ is conditionally Gaussian with mean $\omega'_a \boldsymbol{\mu}_t$ and variance $\omega'_a \boldsymbol{\Omega}_t \omega_a$. We have

$$V_{t+1}(a_0, \mathbf{a}) = a_0(1+r_t) + \omega'_a \mathbf{y}_{t+1} = v(1+r_t) + \omega'_a (\mathbf{y}_{t+1} - (1+r_t)\mathbf{e}) := v(1+r_t) + \omega'_a \mathbf{z}_{t+1}.$$

Note that $\omega'_a \mathbf{z}_{t+1}$ is conditionally Gaussian with mean $\omega'_a \tilde{\boldsymbol{\mu}}_t$, where $\tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t - (1+r_t)\mathbf{e}$, and variance $\omega'_a \boldsymbol{\Omega}_t \omega_a$. We thus have, denoting by Φ the cdf of the standard Gaussian distribution,

$$P_t[V_{t+1}(a_0, \mathbf{a}) \geq x] = P_t[\omega'_a \mathbf{z}_{t+1} \geq x - v(1+r_t)] = 1 - \Phi \left(\frac{x - v(1+r_t) - \omega'_a \tilde{\boldsymbol{\mu}}_t}{\sqrt{\omega'_a \boldsymbol{\Omega}_t \omega_a}} \right)$$

If the right-hand side of (1) holds, we thus have, for all $x \in \mathbb{R}$,

$$\frac{x - v(1+r_t) - \omega'_a \tilde{\boldsymbol{\mu}}_t}{\sqrt{\omega'_a \boldsymbol{\Omega}_t \omega_a}} \leq \frac{x - v(1+r_t) - \omega'_b \tilde{\boldsymbol{\mu}}_t}{\sqrt{\omega'_b \boldsymbol{\Omega}_t \omega_b}}$$

which entails

$$\sqrt{\omega'_a \boldsymbol{\Omega}_t \omega_a} = \sqrt{\omega'_b \boldsymbol{\Omega}_t \omega_b}, \quad \omega'_a \tilde{\boldsymbol{\mu}}_t \geq \omega'_b \tilde{\boldsymbol{\mu}}_t.$$

Thus, for portfolios of given variance, we are led to the mean-variance optimization problem.

- (b) Note that $V_{t+1}(a_0, \mathbf{a})$ is conditionally Gaussian with mean $v(1 + r_t) + \omega'_a \tilde{\mu}_t$ and variance $\omega'_a \Omega_t \omega_a$. For the utility $u(x) = -\exp(-\gamma x)$ we have

$$\begin{aligned} E_t[u\{V_{t+1}(a_0, \mathbf{a})\}] &= -E_t\{-\exp(-\gamma V_{t+1}(a_0, \mathbf{a}))\} \\ &= -\exp\left\{-\gamma v(1 + r_t) - \gamma \omega'_a \tilde{\mu}_t + \frac{\gamma^2}{2} \omega'_a \Omega_t \omega_a\right\} \end{aligned}$$

Maximizing the expected utility is thus equivalent to minimizing

$$f(\omega_a) := -\gamma \omega'_a \tilde{\mu}_t + \frac{\gamma^2}{2} \omega'_a \Omega_t \omega_a$$

At the optimum we have

$$0 = \frac{\partial}{\partial \omega_a} f(\omega_a) = -\gamma \tilde{\mu}_t + \gamma^2 \Omega_t \omega_a.$$

and the solution is

$$\omega_a = \frac{1}{\gamma} \Omega_t^{-1} \tilde{\mu}_t.$$

The coefficient γ is thus the risk aversion of the investor (who chooses to use this utility function). We retrieve the same optimal portfolios as in the mean-variance analysis.

Exercise 2 :

1. We have $\bar{\omega}'_a = \left(\frac{a_1 p_{1t}}{\sum_{j=1}^N a_j p_{jt}}, \dots, \frac{a_N p_{Nt}}{\sum_{j=1}^N a_j p_{jt}} \right)$ (see the course).
2. $E_t(\bar{\omega}'_a \mathbf{y}_{t+1}) = \bar{\omega}'_a \mathbf{c} + \bar{\omega}'_a \mathbf{B} \mathbf{A}$ and $V_t(\bar{\omega}'_a \mathbf{y}_{t+1}) = \bar{\omega}'_a \mathbf{B} \Gamma \mathbf{B}' \bar{\omega}_a + \bar{\omega}'_a \Sigma \bar{\omega}_a$ where the latter term is the idiosyncratic part.
3. (a) It suffices that $\sigma_i^2 < c < \infty$ for all i . Equal weighting with $\bar{\omega}'_a \mathbf{e} = 1$ entails $\bar{\omega}'_a = (1/N, \dots, 1/N)$ which satisfies $\lim_{N \rightarrow \infty} \bar{\omega}'_a \bar{\omega}_a = 0$.
- (b) We have

$$\bar{\omega}'_a \Sigma \bar{\omega}_a = \bar{\omega}'_a \bar{\omega}_a \frac{\bar{\omega}'_a \Sigma \bar{\omega}_a}{\bar{\omega}'_a \bar{\omega}_a} \leq \rho(\Sigma) \bar{\omega}'_a \bar{\omega}_a \leq c \bar{\omega}'_a \bar{\omega}_a \rightarrow 0.$$

- (c) $\bar{\omega}'_a \Sigma \bar{\omega}_a = \frac{1}{N^2} \sum_{i,j=1}^N \rho^{|i-j|} \leq \frac{2}{N^2} \sum_{i=1}^N \sum_{j=0}^{\infty} \rho^j = \frac{2}{N} \frac{1}{1-\rho} \rightarrow 0$ as $N \rightarrow \infty$.
- (d) We have

$$\text{Cov}_t(\bar{\omega}'_a \mathbf{y}_{t+1}, \bar{\omega}'_b \mathbf{y}_{t+1}) = \bar{\omega}'_a \mathbf{B} \Gamma \mathbf{B}' \bar{\omega}_b + \bar{\omega}'_a \Sigma \bar{\omega}_b$$

where the last term tends to 0 by the Cauchy-Schwarz inequality ($\bar{\omega}'_a \Sigma \bar{\omega}_b \leq \{\bar{\omega}'_a \Sigma \bar{\omega}_a\}^{1/2} \{\bar{\omega}'_b \Sigma \bar{\omega}_b\}^{1/2}$). Hence

$$\text{Cov}_t(\bar{\omega}'_a \mathbf{y}_{t+1}, \bar{\omega}'_b \mathbf{y}_{t+1}) \approx \lim_{N \rightarrow \infty} \text{Cov}_t(\bar{\omega}'_a \mathbf{y}_{t+1}, \bar{\omega}'_b \mathbf{y}_{t+1}) = \bar{\omega}'_a \mathbf{B} \Gamma \mathbf{B}' \bar{\omega}_b.$$

This limit does not vanish when $\bar{\omega}'_a \bar{\omega}_b = 0$.



ENSAE - 3rd year, Master 2
January 2022

FINANCIAL ECONOMETRICS
Closed book examination, no calculator - 2 hours

Notations reminder : Consider an economy with N risky assets and one risk-free asset. Let $p_{i,t}$ the price of asset i at time t ($i = 1, \dots, N$) and let $y_{i,t+1} = \frac{p_{i,t+1}}{p_{i,t}}$ and $\mathbf{y}_{t+1} = (y_{1,t+1}, \dots, y_{N,t+1})'$. The risk-free asset has price 1 at time t and return r_t between time t and $t + 1$.

The conditional moments of $\mathbf{y}_{t+1}$ (given the available information at time t) are :

$$E_t(\mathbf{y}_{t+1}) = \boldsymbol{\mu}_t, \quad \text{Var}_t(\mathbf{y}_{t+1}) = \boldsymbol{\Omega}_t$$

where $\boldsymbol{\mu}_t \in \mathbb{R}^N$ and $\boldsymbol{\Omega}_t$ is a positive-definite matrix (denoted $\boldsymbol{\Omega}$ when independent of t).

Let $\mathbf{z}_{t+1} = (z_{i,t+1})_{i=1,\dots,N}$ and $\tilde{\boldsymbol{\mu}}_t = (\tilde{\mu}_{i,t})_{i=1,\dots,N}$ the vectors

$$\mathbf{z}_{t+1} = \mathbf{y}_{t+1} - (1 + r_t)\mathbf{e}, \quad \tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t - (1 + r_t)\mathbf{e}, \quad \text{where } \mathbf{e} = (1, \dots, 1)'$$

Denote by V_t the value of a portfolio composed, at time t , of a_0 units of the risk-free asset and a_i units of assets i , for $i = 1, \dots, N$: $V_t = a_0 + \boldsymbol{\omega}'_a \mathbf{e}$ where $\boldsymbol{\omega}_a = (a_1 p_{1,t}, \dots, a_N p_{N,t})'$ and $\mathbf{a} = (a_1, \dots, a_N)'$. Denote by $z_{t+1}(\mathbf{a})$ the net excess return of the risky part of the portfolio : $z_{t+1}(\mathbf{a}) = \frac{\mathbf{a}' \mathbf{p}_{t+1}}{\mathbf{a}' \mathbf{p}_t} - (1 + r_t)$ where $\mathbf{p}_t = (p_{1t}, \dots, p_{Nt})'$.

Problem : Suppose that the net returns satisfy the model

$$z_{i,t} = c_i + \beta_i F_t + \epsilon_{i,t}, \quad i = 1, \dots, N, \quad t = 1, \dots, n$$

where F_t is a common factor, $F_t \in \mathbb{R}$, the c_i 's and β_i 's are parameters, $c_i, \beta_i \in \mathbb{R}$. Let $\mathbf{c} = (c_1, \dots, c_N)'$ and $\boldsymbol{\epsilon}_t = (\epsilon_{1,t}, \dots, \epsilon_{N,t})'$. The sequences $(F_t)_{t=1,\dots,n}$ and $(\boldsymbol{\epsilon}_t)$ are assumed independent and identically distributed (i.i.d.), independent from each other, and such that $F_t \sim \mathcal{N}(m, \eta^2)$, $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_N)$ where $\mathbf{I}_N$ is the N -dimensional identity matrix, $\eta^2, \sigma^2 > 0$. Unless otherwise stated, the factor F_t is assumed to be observable at time t .

1. Write the model in vector form. Compute $\tilde{\boldsymbol{\mu}}_t$ and show that $\boldsymbol{\Omega} = \boldsymbol{\beta} \boldsymbol{\beta}' \eta^2 + \sigma^2 \mathbf{I}_N$ where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_N)'$. Give, for each risky asset, the systemic risk and the idiosyncratic risk.

2. Show that, for some constant τ^2 depending on η^2, σ^2 and $\boldsymbol{\beta}$,

$$\boldsymbol{\Omega}^{-1} = \frac{1}{\sigma^2} \left(\mathbf{I}_N - \frac{\boldsymbol{\beta}\boldsymbol{\beta}'}{\boldsymbol{\beta}'\boldsymbol{\beta}} \right) + \frac{1}{\tau^2} \frac{\boldsymbol{\beta}\boldsymbol{\beta}'}{\boldsymbol{\beta}'\boldsymbol{\beta}}.$$

3. Deduce that the composition in risky assets of the efficient portfolios belongs to the space generated by the vectors $\boldsymbol{\beta}$ and $\mathbf{c}$.
4. Show that the composition of efficient portfolios is proportional to the vector $\boldsymbol{\beta}$ when $\mathbf{c} = 0$ and compute their performance (defined by the Sharpe ratio S_N) in this case.
5. Consider the portfolios such that $\text{Cov}_t(z_{t+1}(\mathbf{a}), F_{t+1}) = 0$. Among these portfolios, which have the best performance?
6. Can the model be used to test the CAPM? the APT?
7. In this question, assume that $N \rightarrow \infty$ and that $\mathbf{c} = 0$. Let

$$\bar{\beta}_N = \frac{1}{N} \sum_{i=1}^N \beta_i, \quad \sigma_{\beta, N}^2 = \frac{1}{N} \sum_{i=1}^N (\beta_i - \bar{\beta}_N)^2$$

and suppose that $\lim_{N \rightarrow \infty} \bar{\beta}_N = 1$ and $\lim_{N \rightarrow \infty} \sigma_{\beta, N}^2 = \sigma_{\beta, \infty}^2 > 0$.

- (a) Show that the composition in asset j of the efficient portfolios tends to 0, for any j (the proportionality factor of question 3 will be assumed to be independent of N).
- (b) Compute $\lim_{N \rightarrow \infty} S_N$ and interpret this quantity in terms of risk diversification.
8. We have observations of the net returns, $\mathbf{z}_1, \dots, \mathbf{z}_n$, and observations of the factor $f_1, \dots, f_n$. Derive a consistent estimator of $\boldsymbol{\beta}$.
9. If now the factor is non observable, show that it is not restrictive to assume $\eta^2 = 1$. Propose an estimator of $\boldsymbol{\beta}$ based on the spectral decomposition of an estimator of $\boldsymbol{\Omega}$.

SOLUTION

(NB : Not all questions are corrected in details)

1. We have $\mathbf{z}_{t+1} = \mathbf{c} + \beta F_{t+1} + \epsilon_{t+1}$. Thus $\tilde{\boldsymbol{\mu}}_t = \mathbf{c} + \beta m$.

The formula for $\boldsymbol{\Omega}$ is obtained using the independence between F_t and ϵ_{it} . The variance for the asset i has the decomposition

$$\Omega_{ii} = \beta_i^2 \eta^2 + \sigma^2,$$

where the first term is the systemic part of risk and the second term is the idiosyncratic part.

2. We get $\tau^2 = \sigma^2 + \eta^2 \beta' \beta$ by computing the product

$$\boldsymbol{\Omega}^{-1} \boldsymbol{\Omega} = \left\{ \frac{1}{\sigma^2} \left(\mathbf{I}_N - \frac{\beta \beta'}{\beta' \beta} \right) + \frac{1}{\tau^2} \frac{\beta \beta'}{\beta' \beta} \right\} (\beta \beta' \eta^2 + \sigma^2 \mathbf{I}_N)$$

and by noting that $(\beta \beta')^2 = \beta (\beta' \beta) \beta' = (\beta' \beta) \beta \beta'$ since $\beta' \beta$ is a scalar.

3. Optimal portfolios have a composition in risky assets (at time t) given by

$$\boldsymbol{\omega} = (a_1 p_{1,t}, \dots, a_N p_{N,t})' = \frac{1}{\lambda} \boldsymbol{\Omega}^{-1} \tilde{\boldsymbol{\mu}}_t$$

where λ is a risk aversion coefficient, that is

$$\boldsymbol{\omega} = \frac{1}{\lambda} \left[\frac{\mathbf{c}}{\sigma^2} + \left\{ \left(\frac{1}{\tau^2} - \frac{1}{\sigma^2} \right) \frac{\beta' \mathbf{c}}{\beta' \beta} + \frac{m}{\tau^2} \right\} \beta \right].$$

Hence, the efficient portfolios belong to the space generated by the vectors β and $\mathbf{c}$.

4. If $\mathbf{c} = 0$ we get the vectors that are colinear to β . The performance of the efficient portfolios coincide with the market performance, given by

$$S_N^2 = \tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}^{-1} \tilde{\boldsymbol{\mu}}_t = \frac{m^2 \beta' \beta}{\sigma^2 + \eta^2 \beta' \beta} = \frac{m^2}{\tau^2} \beta' \beta.$$

5. The net excess return of the risky part of the portfolio is

$$z_{t+1}(\mathbf{a}) = \frac{\mathbf{a}' \mathbf{p}_{t+1}}{\mathbf{a}' \mathbf{p}_t} - (1 + r_t) = \bar{\boldsymbol{\omega}}_a' \mathbf{z}_{t+1}.$$

Thus

$$\text{Cov}(z_{t+1}(\mathbf{a}), F_{t+1}) = \bar{\boldsymbol{\omega}}_a' \text{Cov}(\mathbf{z}_{t+1}, F_{t+1}) = \bar{\boldsymbol{\omega}}_a' \beta \eta^2,$$

which is equal to 0 iff $\bar{\omega}'_a \beta = 0$. For such portfolios we have $E_t\{z_{t+1}(\mathbf{a})\} = \bar{\omega}'_a \mathbf{c}$ and $\text{Var}_t\{z_{t+1}(\mathbf{a})\} = \bar{\omega}'_a \text{Var}_t\{\mathbf{z}_{t+1}\} \bar{\omega}_a = \bar{\omega}'_a \bar{\omega}_a \sigma^2$. Among these portfolios, the most performant are those maximizing the Sharpe ratio

$$\frac{E_t\{z_{t+1}(\mathbf{a})\}}{\sqrt{\text{Var}_t\{z_{t+1}(\mathbf{a})\}}} = \frac{\bar{\omega}'_a \mathbf{c}}{\{\bar{\omega}'_a \bar{\omega}_a\}^{1/2} \sigma},$$

that is the portfolios such that $\bar{\omega}_a = K \mathbf{c}$ with $K > 0$ such that $\bar{\omega}'_a \mathbf{e} = 1$.

6. The CAPM is based on the choice of the net return of the market portfolio for the factor F_t . This theory can be tested by testing the hypothesis $\mathbf{c} = 0$ in the model $\mathbf{z}_{t+1} = \mathbf{c} + \beta F_t + \epsilon_t$.

In the APT theory, F_t can be any factor. The absence of arbitrage opportunity entails (when σ^2 is negligible) by $\tilde{\mu}_t = \mathbf{B} \lambda$ where $\mathbf{B}$ is the $N \times K$ matrix of factor loadings (here $K = 1$ and $\mathbf{B} = \beta$) and λ is a vector of size K (here a scalar). Thus we must also have $\mathbf{c} = 0$ or $\mathbf{c}$ proportional to β for the APT.

7. In view of question 3, the efficient portfolios are proportional to

$$\frac{m}{\tau^2} \beta = \frac{m}{\sigma^2 + \eta^2 \beta' \beta} \beta.$$

For i fixed, $\frac{m \beta_i}{\sigma^2 + \eta^2 \beta' \beta}$ tends to 0, because

$$\beta' \beta = \sum_{j=1}^N \beta_j^2 = N(\sigma_{\beta, N}^2 + \bar{\beta}_N^2) \sim N(\sigma_{\beta, \infty}^2 + 1).$$

We also have

$$S_N^2 \sim \frac{m^2}{\eta^2},$$

which means that the market performance reduces to the Sharpe performance of F_t (even if F_t is not necessarily the return of a portfolio!). We thus have total diversification of the idiosyncratic risk when $N \rightarrow \infty$.

8. The GLS estimator of β_i is obtained by LS estimation equation by equation :

$$\hat{\beta}_i = \frac{\sum_{t=1}^n (z_{it} - \mu_{in})(F_t - \hat{F}_n)}{\sum_{t=1}^n (F_t - \hat{F}_n)^2},$$

where

$$\mu_{in} = \frac{1}{n} \sum_{t=1}^n z_{it}, \quad \hat{F}_n = \frac{1}{n} \sum_{t=1}^n F_t.$$

A consistent estimator of β is thus

$$\hat{\beta} = \frac{\sum_{t=1}^n (\mathbf{z}_t - \mu_n)(F_t - \hat{F}_n)}{\sum_{t=1}^n (F_t - \hat{F}_n)^2}.$$

9. One can assume $\eta^2 = 1$: otherwise F_t could be replaced by $F_t^* = F_t/\eta^2$ and $\boldsymbol{\beta}$ by $\boldsymbol{\beta}^* = \boldsymbol{\beta}\eta^2$. We thus have $\boldsymbol{\Omega} = \boldsymbol{\beta}\boldsymbol{\beta}' + \sigma^2\mathbf{I}_N$ The vector $\boldsymbol{\beta}$ can be estimated by $\sqrt{\hat{\lambda}_1}\hat{\mathbf{e}}_1$ where $\hat{\lambda}_1$ is the largest eigenvalue of $\hat{\boldsymbol{\Omega}}$ (empirical variance of the $\mathbf{z}_t$, $t = 1, \dots, n$) and $\hat{\mathbf{e}}_1$ is the corresponding eigenvector with norm 1.

FINANCIAL ECONOMETRICS
Closed book examination, no calculator - 2 hours

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The conditional moments of $\mathbf{y}_{t+1}$ (given the available information at time t) are :

$$E_t(\mathbf{y}_{t+1}) = \boldsymbol{\mu}_t, \quad \text{Var}_t(\mathbf{y}_{t+1}) = \boldsymbol{\Omega}_t$$

where $\boldsymbol{\mu}_t \in \mathbb{R}^N$ and $\boldsymbol{\Omega}_t$ is a positive-definite matrix (denoted $\boldsymbol{\Omega}$ when independent of t).

Let $\mathbf{z}_{t+1} = (z_{i,t+1})_{i=1,\dots,N}$ and $\tilde{\boldsymbol{\mu}}_t = (\tilde{\mu}_{i,t})_{i=1,\dots,N}$ the vectors

$$\mathbf{z}_{t+1} = \mathbf{y}_{t+1} - (1 + r_t)\mathbf{e}, \quad \tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t - (1 + r_t)\mathbf{e}, \quad \text{where } \mathbf{e} = (1, \dots, 1)'$$

Denote by V_t the value of a portfolio composed, at time t , of a_0 units of the risk-free asset and a_i units of assets i , for $i = 1, \dots, N$: $V_t = V_t(a_0, \mathbf{a}) = a_0 + \boldsymbol{\omega}'_t \mathbf{a}$ where $\boldsymbol{\omega}_t = (a_1 p_{1,t}, \dots, a_N p_{N,t})'$ and $\mathbf{a} = (a_1, \dots, a_N)'$. Denote by $z_{t+1}(\mathbf{a})$ the net excess return of the risky part of the portfolio : $z_{t+1}(\mathbf{a}) = \frac{\mathbf{a}' \mathbf{p}_{t+1}}{\mathbf{a}' \mathbf{p}_t} - (1 + r_t)$ where $\mathbf{p}_t = (p_{1,t}, \dots, p_{N,t})'$.

Exercise 1 : Suppose the risky assets are gathered by K subsets of assets with similar characteristics. An investor wishes to build optimal portfolios under the constraint that the portfolio at time t is equally invested in assets of the same subset. For instance if Assets 1 and 2 belong to the same subset, the investor will choose portfolios such that $a_1 p_{1,t} = a_2 p_{2,t}$. More generally, the constraint can be expressed as $\mathbf{A} \boldsymbol{\omega}_t = \mathbf{0}_{M \times 1}$ where $\mathbf{A}$ is a $M \times N$ matrix, and $M < N$ is the number of equality constraints. It can be assumed that $\mathbf{A}$ has rank M .

1. Find a matrix A corresponding to $K = 1$, that is $a_i p_{i,t} = a_j p_{j,t}$ for all i and j .
2. The investor is looking for optimal portfolios, in the mean-variance sense, which also satisfy the constraint previously defined over the K subsets. Therefore, for portfolios of initial value $V_t(a_0, \mathbf{a}) = v$, he/she wishes to solve

$$\begin{cases} \max_{a_0, \mathbf{a}} E_t\{V_{t+1}(a_0, \mathbf{a})\} \\ \text{under the constraints} \quad \text{Var}_t\{V_{t+1}(a_0, \mathbf{a})\} = \sigma_t^2, \quad \mathbf{A} \boldsymbol{\omega}_t = \mathbf{0}_{M \times 1}, \end{cases}$$

where σ_t^2 is a given variance level.

(a) Prove that the composition in risky assets is obtained by solving

$$\begin{cases} \max_{\omega_{\mathbf{a}}} \omega'_{\mathbf{a}} \tilde{\mu}_t \\ \text{u.c.} & \omega'_{\mathbf{a}} \Omega_t \omega_{\mathbf{a}} = \sigma_t^2, \quad \mathbf{A} \omega_{\mathbf{a}} = \mathbf{0}_{M \times 1}. \end{cases}$$

(b) Write the Lagrangian and the first-order conditions.

(c) Show that the solutions have the form

$$\omega_{\mathbf{a}}^* = \frac{1}{\lambda} \Omega_t^{-1} \{ \mathbf{I} - \mathbf{A}' (\mathbf{A} \Omega_t^{-1} \mathbf{A}')^{-1} \mathbf{A} \Omega_t^{-1} \} \tilde{\mu}_t, \quad (1)$$

where $\mathbf{I}$ is the $N \times N$ identity matrix.

(d) How is λ related to σ_t^2 ? show that $\lambda > 0$ and give an interpretation of λ .

3. Graph the efficient frontier corresponding to portfolios defined by (1) by comparison with the usual efficient frontier.

Exercise 2 : Suppose that the net returns satisfy the model

$$z_{i,t+1} = \beta_{it} z_{t+1}^{(m)} + \epsilon_{i,t+1}, \quad i = 1, \dots, N, \quad t = 1, \dots, n,$$

where $z_{t+1}^{(m)}$ is the net return of the market portfolio and, letting $\epsilon_{t+1} = (\epsilon_{1,t+1}, \dots, \epsilon_{N,t+1})$ and Σ denoting a $N \times N$ positive-definite matrix,

$$E_t(\epsilon_{t+1}) = 0, \quad E_t(\epsilon_{t+1} \epsilon'_{t+1}) = \Sigma, \quad \text{Cov}_t(\epsilon_{t+1}, z_{t+1}^{(m)}) = \mathbf{0}.$$

Let β_t denote the vector with entries β_{it} .

1. Give an interpretation of this model. In particular, how is it related to the CAPM theory? Recall the assumptions on which this theory relies.
2. Justify the relations

$$\tilde{\mu}_t = \beta_t E_t(z_{t+1}^{(m)}), \quad \Omega_t = \beta_t \text{Var}_t(z_{t+1}^{(m)}) \beta'_t + \Sigma, \quad \beta_t = \frac{\text{Cov}_t(\mathbf{z}_{t+1}, z_{t+1}^{(m)})}{\text{Var}_t(z_{t+1}^{(m)})}.$$

3. We have observations $\mathbf{z}_1, \dots, \mathbf{z}_n$ and $z_1^{(m)}, \dots, z_n^{(m)}$. We assume that $\beta_t = \beta$ is independent of t . What is the Generalized Least Squares (GLS) estimator $\hat{\beta}_n$ of β ?
4. Assume that

$$\sqrt{n} (\hat{\beta}_n - \beta) \xrightarrow{d} \mathcal{N} \left(0, \frac{1}{\sigma_m^2} \Sigma \right)$$

where $\sigma_m^2 = \text{Var}(z_{t+1}^{(m)})$.

We wish to test the hypothesis that the assets 1 and 2 have same risk. Propose a test.

SOLUTION

(NB : Not all questions are corrected in details)

Exercise 1 :

1. A $(N - 1) \times N$ matrix $\mathbf{A}$ corresponding to the constraints is

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 & \dots & 0 \\ 0 & 1 & -1 & 0 & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \dots & & 1 & -1 \end{pmatrix}$$

2. (a) See the slides of Chapter 2.
 (b) Introducing the Lagrangian function, the programme writes

$$\begin{cases} \max_{\omega_{\mathbf{a}}, \lambda, \mathbf{\Lambda}} \mathcal{L}(\omega_{\mathbf{a}}, \lambda, \mathbf{\Lambda}) \\ \mathcal{L}(\omega_{\mathbf{a}}, \lambda, \mathbf{\Lambda}) = \omega'_{\mathbf{a}} \tilde{\mu}_t - \frac{\lambda}{2} \{ \omega'_{\mathbf{a}} \mathbf{\Omega}_t \omega_{\mathbf{a}} - \sigma_t^2 \} - \mathbf{\Lambda}' \mathbf{A} \omega_{\mathbf{a}}, \end{cases}$$

where λ is a real and $\mathbf{\Lambda}' = (\lambda_1, \dots, \lambda_M)$ is a $1 \times M$ vector corresponding to the M constraints. The first-order conditions are

$$\begin{cases} \frac{\partial \mathcal{L}(\omega_{\mathbf{a}}, \lambda, \mathbf{\Lambda})}{\partial \omega_{\mathbf{a}}} = \tilde{\mu}_t - \lambda \mathbf{\Omega}_t \omega_{\mathbf{a}} - \mathbf{A}' \mathbf{\Lambda} = 0, \\ \frac{\partial \mathcal{L}(\omega_{\mathbf{a}}, \lambda, \mathbf{\Lambda})}{\partial \lambda} = -\frac{1}{2} \{ \omega'_{\mathbf{a}} \mathbf{\Omega}_t \omega_{\mathbf{a}} - \sigma_t^2 \} = 0, \quad \frac{\partial \mathcal{L}(\omega_{\mathbf{a}}, \lambda)}{\partial \mathbf{\Lambda}} = \mathbf{A} \omega_{\mathbf{a}} = \mathbf{0}. \end{cases}$$

- (c) From the first condition we deduce $\omega_{\mathbf{a}}^* = \frac{1}{\lambda} \mathbf{\Omega}_t^{-1} \{ \tilde{\mu}_t - \mathbf{A}' \mathbf{\Lambda} \}$ and the last condition gives $\mathbf{A} \mathbf{\Omega}_t^{-1} \{ \tilde{\mu}_t - \mathbf{A}' \mathbf{\Lambda} \} = \mathbf{0}$. We deduce $\mathbf{\Lambda} = (\mathbf{A} \mathbf{\Omega}_t^{-1} \mathbf{A}')^{-1} \mathbf{A} \mathbf{\Omega}_t^{-1} \tilde{\mu}_t$, noting that $\mathbf{A} \mathbf{\Omega}_t^{-1} \mathbf{A}'$ is nonsingular because $\mathbf{A}$ has full-row rank and $\mathbf{\Omega}_t^{-1}$ is positive definite. Thus

$$\omega_{\mathbf{a}}^* = \frac{1}{\lambda} \mathbf{\Omega}_t^{-1} (\mathbf{I} - \mathbf{B}_t) \tilde{\mu}_t, \quad \mathbf{B}_t = \mathbf{A}' (\mathbf{A} \mathbf{\Omega}_t^{-1} \mathbf{A}')^{-1} \mathbf{A} \mathbf{\Omega}_t^{-1}$$

- (d) We also have, in view of the second condition, $\omega_{\mathbf{a}}^{*'} \mathbf{\Omega}_t \omega_{\mathbf{a}}^* = \sigma_t^2$, hence

$$\sigma_t^2 = \frac{1}{\lambda^2} \tilde{\mu}_t' (\mathbf{I} - \mathbf{B}_t)' \mathbf{\Omega}_t^{-1} (\mathbf{I} - \mathbf{B}_t) \tilde{\mu}_t.$$

Using the expression of $\mathbf{B}_t$, it can be checked that

$$(\mathbf{I} - \mathbf{B}_t)' \mathbf{\Omega}_t^{-1} (\mathbf{I} - \mathbf{B}_t) = (\mathbf{I} - \mathbf{B}_t)' \mathbf{\Omega}_t^{-1}.$$

Hence,

$$\sigma_t^2 = \frac{1}{\lambda^2} \tilde{\boldsymbol{\mu}}_t' (\mathbf{I} - \mathbf{B}_t') \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t.$$

For the optimal portfolio we have

$$m_t = E_t(V_{t+1}) = v(1 + r_t) + \boldsymbol{\omega}'_{\mathbf{a}} \tilde{\boldsymbol{\mu}}_t = v(1 + r_t) + \frac{1}{\lambda} \tilde{\boldsymbol{\mu}}_t' (\mathbf{I} - \mathbf{B}_t')' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t.$$

In view of the expression of σ_t^2 , we have $\tilde{\boldsymbol{\mu}}_t' (\mathbf{I} - \mathbf{B}_t')' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t > 0$. Thus λ has to be positive to ensure a better expected return than if the portfolio was fully invested in the riskless asset.

3. The efficient frontier is obtained by noting that

$$\sigma_t^2 = \frac{\{m_t - v(1 + r_t)\}^2}{\tilde{\boldsymbol{\mu}}_t' (\mathbf{I} - \mathbf{B}_t')' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t}.$$

It is a semi-parabola in the plan (σ_t, m_t) , which intersects the vertical axis at $v(1 + r_t)$. The curve is below the usual efficient frontier because $\tilde{\boldsymbol{\mu}}_t' \mathbf{B}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t \geq 0$. The two curves coincide when $\mathbf{A} \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t = 0$.

Exercise 2 :

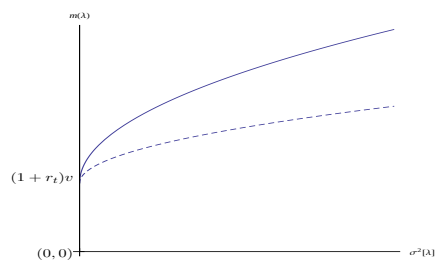
1. A consequence of the CAPM theory is that, in the regression of $\mathbf{z}_{t+1}$ on the net returns $z_{t+1}^{(m)}$ of the market portfolio, the intercept α is equal to 0. The model is thus compatible with this theory. See the slides for the hypotheses of the CAPM.
2. See the slides.
3. We know that the GLS estimator $\hat{\boldsymbol{\beta}}_n = (\hat{\beta}_{1n}, \dots, \hat{\beta}_{Nn})'$ of $\boldsymbol{\beta}$ coincides with the LS estimator whose components are obtained equation by equation. Thus

$$\hat{\beta}_{in} = \frac{\frac{1}{n} \sum_{t=1}^n z_{i,t+1} z_{t+1}^{(m)}}{\frac{1}{n} \sum_{t=1}^n (z_{t+1}^{(m)})^2}, \quad i = 1, \dots, N.$$

4. The hypothesis writes $H_0 : \mathbf{R} \boldsymbol{\beta} = 0$ where $\mathbf{R} = (1, -1, 0, \dots)$. Under H_0 we have $\sqrt{n} \mathbf{R} \hat{\boldsymbol{\beta}}_n \xrightarrow{d} \mathcal{N} \left(0, \frac{1}{\sigma_m^2} \mathbf{R} \boldsymbol{\Sigma} \mathbf{R}' \right)$. Hence, under H_0 ,

$$n \mathbf{R} \hat{\boldsymbol{\beta}}_n \left(\frac{1}{\sigma_m^2} \mathbf{R} \boldsymbol{\Sigma} \mathbf{R}' \right)^{-1} \hat{\boldsymbol{\beta}}_n' \mathbf{R}' \xrightarrow{d} \chi^2(1).$$

A test statistic is thus $T_n = n \mathbf{R} \hat{\boldsymbol{\beta}}_n \left(\frac{1}{\hat{\sigma}_m^2} \mathbf{R} \hat{\boldsymbol{\Sigma}} \mathbf{R}' \right)^{-1} \hat{\boldsymbol{\beta}}_n' \mathbf{R}'$ where $\hat{\boldsymbol{\Sigma}}$ is a consistent estimator of $\boldsymbol{\Sigma}$ and $\hat{\sigma}_m^2$ is a consistent estimator of σ_m^2 . We can take $\hat{\boldsymbol{\Sigma}} = \frac{1}{n} \sum_{t=1}^n \hat{\boldsymbol{\epsilon}}_t \hat{\boldsymbol{\epsilon}}_t'$ where the $\hat{\boldsymbol{\epsilon}}_t$'s are the residuals of the regression, and $\hat{\sigma}_m^2$ equal to the empirical variance of the $z_{t+1}^{(m)}$'s. Under H_0 , $T_n \xrightarrow{d} \chi^2(1)$.





3A/Master/MS - first session
2023-24 - Semester 1

FINANCIAL ECONOMETRICS

Closed book examination, no calculator - 2 hours

Notations reminder : Consider an economy with N risky assets and one risk-free asset. Let $p_{i,t}$ the price of asset i at time t ($i = 1, \dots, N$) and let $y_{i,t+1} = \frac{p_{i,t+1}}{p_{i,t}}$ and $\mathbf{y}_{t+1} = (y_{1,t+1}, \dots, y_{N,t+1})'$. The risk-free asset has price 1 at time t and return r_t between time t and $t + 1$.

The conditional moments of $\mathbf{y}_{t+1}$ (given the available information at time t) are :

$$E_t(\mathbf{y}_{t+1}) = \boldsymbol{\mu}_t, \quad \text{Var}_t(\mathbf{y}_{t+1}) = \boldsymbol{\Omega}_t$$

where $\boldsymbol{\mu}_t \in \mathbb{R}^N$ and $\boldsymbol{\Omega}_t$ is a positive-definite matrix (denoted $\boldsymbol{\Omega}$ when independent of t).

Let $\mathbf{z}_{t+1} = (z_{i,t+1})_{i=1,\dots,N}$ and $\tilde{\boldsymbol{\mu}}_t = (\tilde{\mu}_{i,t})_{i=1,\dots,N}$ the vectors

$$\mathbf{z}_{t+1} = \mathbf{y}_{t+1} - (1 + r_t)\mathbf{e}, \quad \tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t - (1 + r_t)\mathbf{e}, \quad \text{where } \mathbf{e} = (1, \dots, 1)'$$

Denote by V_t the value of a portfolio composed, at time t , of a_0 units of the risk-free asset and a_i units of assets i , for $i = 1, \dots, N$: $V_t = a_0 + \boldsymbol{\omega}'_a \mathbf{e}$ where $\boldsymbol{\omega}_a = (a_1 p_{1,t}, \dots, a_N p_{N,t})'$ and $\mathbf{a} = (a_1, \dots, a_N)'$. Denote by $z_{t+1}(\mathbf{a})$ the net excess return of the risky part of the portfolio : $z_{t+1}(\mathbf{a}) = \frac{\mathbf{a}' \mathbf{p}_{t+1}}{\mathbf{a}' \mathbf{p}_t} - (1 + r_t)$ where $\mathbf{p}_t = (p_{1t}, \dots, p_{Nt})'$.

The exercises are independent.

Exercise 1 (7 points) :

1. Show that $z_{t+1}(\mathbf{a}) = \bar{\boldsymbol{\omega}}'_a \mathbf{z}_{t+1}$, where $\bar{\boldsymbol{\omega}}_a = \frac{\boldsymbol{\omega}_a}{\boldsymbol{\omega}'_a \mathbf{e}}$.
2. We look for portfolios whose net excess returns have minimal variance, i.e. the solutions of the program

$$\begin{cases} \min_{\mathbf{a}} \text{Var}_t(z_{t+1}(\mathbf{a})) \\ \text{u.c. } \bar{\boldsymbol{\omega}}'_a \mathbf{e} = 1. \end{cases}$$

- (a) Compare this approach with the classical mean-variance approach of Markowitz. Solve the program.
- (b) Show that efficient portfolios in the Markowitz sense are not solutions, except when the gross returns of the risky assets have the same conditional mean.
- (c) Plot the solutions on the mean-variance graph.

Exercise 2 (8 points) Suppose that the returns satisfy the model

$$\mathbf{y}_t = \mathbf{c} + \mathbf{B}\mathbf{f}_t + \mathbf{u}_t, \quad E(\mathbf{u}_t | \mathbf{f}_t) = \mathbf{0}, \quad E(\mathbf{u}_t \mathbf{u}_t' | \mathbf{f}_t) = \mathbf{\Sigma}$$

where $\mathbf{f}_t$ is a common factor, $\mathbf{f}_t \in \mathbb{R}^K$, $\mathbf{c} \in \mathbb{R}^N$, $\mathbf{B}$ is $N \times K$ matrix of coefficients, $\mathbf{\Sigma}$ is a positive definite matrix. Assume $E(\mathbf{f}_t) = \mathbf{0}$ and $\text{Var}(\mathbf{f}_t) = \mathbf{\Gamma}$.

1. In the particular case of the Fama-French model, specify the factors and interpret them. Is this model a dynamic or a static factor model?
2. Compute $E(\mathbf{y}_t)$ and $\text{Var}(\mathbf{y}_t)$. Show that the latter matrix is non-singular.

3. Assume $\mathbf{c} = \mathbf{0}$. We have observations $\mathbf{y}_1, \dots, \mathbf{y}_n, \mathbf{f}_1, \dots, \mathbf{f}_n$. Let $\mathbf{Y} = \begin{pmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_n \end{pmatrix} \in \mathbb{R}^{nN}$.

- (a) Show that the model can be written under the form $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U}$ where $\mathbf{X}$ is a matrix and $\boldsymbol{\beta} \in \mathbb{R}^{NK}$ a vector of parameters. *Hint : Apply the vec operator to the equation of $\mathbf{y}_t$ and use $\text{vec}(ABC) = (C' \otimes A)\text{vec}(B)$.*
- (b) Deduce the Generalized Least Squares (GLS) estimator of $\boldsymbol{\beta}$,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\boldsymbol{\Sigma}_U^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Sigma}_U^{-1}\mathbf{Y},$$

where $\boldsymbol{\Sigma}_U$ is the variance of $\mathbf{U}$, and show that it coincides with the LS estimator.

- (c) Deduce the GLS estimator of $\mathbf{B}$ in function of the $\mathbf{y}_t$'s and $\mathbf{f}_t$'s.

Exercise 3 (5 points) : We want to test the validity of the CAPM in an economy composed of $N = 6$ risky assets. Table 1 provides the estimated regression coefficients α_i and β_i for the 6 stocks, the estimated variances σ_ϵ^2 of the error terms, the estimated Sharpe ratios. The variance of the market portfolio is denoted σ_m^2 . Comment on the results.

TABLE 1 – Estimation results for 6 stocks.							
Assets	1	2	3	4	5	6	Market portfolio
$\alpha_i \times 10^3$	0.84	0.71	0.23	-0.41	-4.28	-1.21	-
p -value for $\alpha_i = 0$	0.72	0.76	0.04	0.01	0.52	0.06	-
β_i	1.23	1.20	0.52	1.44	2.28	0.46	-
$\beta_i^2 \sigma_m^2 \times 10^4$	5.77	5.48	1.04	7.91	19.8	0.80	-
$\sigma_\epsilon^2 \times 10^4$	3.50	4.18	4.22	12.0	26.7	4.64	-
Sharpe $\times 10^2$	-9.49	-10.36	-1.05	-12.1	-13.2	-26.4	-10.6
Wald test statistic value of the CAPM : $W_n = 17.2$.							

TABLE 2 – Quantiles of the $\chi^2(6)$ distribution.						
α	0.99	0.95	0.90	0.10	0.05	0.01
$\chi_6^2(1 - \alpha)$	0.87	1.64	2.20	10.64	12.59	16.81

SOLUTION

(NB : Not all questions are corrected in details)

Exercise 1 :

1. See the course. When the a_i are positive we can interpret $\bar{\omega}_{\mathbf{a}}$ as a vector of weights of the risky assets.
2. (a) In the mean-variance approach, one has to minimize the conditional variance $\omega'_{\mathbf{a}} \Omega_t \omega_{\mathbf{a}}$ with respect to $\omega_{\mathbf{a}}$, under the constraint that the conditional mean is fixed. The number of units in the risk-free asset is used to satisfy the budget constraint and the composition in risky assets is determined up to a multiplicative constant (related to the risk aversion of the investor). In the present approach we minimize $\text{Var}_t(z_{t+1}(\mathbf{a})) = \bar{\omega}'_{\mathbf{a}} \Omega_t \bar{\omega}_{\mathbf{a}}$ under the constraint of using weights (when the a_i are positive).

Introducing the Lagrangian function, the programme writes

$$\begin{cases} \min_{\bar{\omega}_{\mathbf{a}}, \lambda} \mathcal{L}(\bar{\omega}_{\mathbf{a}}, \lambda) \\ \mathcal{L}(\bar{\omega}_{\mathbf{a}}, \lambda) = \bar{\omega}'_{\mathbf{a}} \Omega_t \bar{\omega}_{\mathbf{a}} - 2\lambda\{\bar{\omega}'_{\mathbf{a}} e - 1\}, \end{cases}$$

where λ is a real. From the first-order conditions, we get the solution

$$\bar{\omega}_{\mathbf{a}} = \frac{\Omega_t^{-1} e}{e' \Omega_t^{-1} e}$$

which corresponds to the compositions in units of assets

$$\omega_{\mathbf{a}} = \lambda \Omega_t^{-1} e, \quad \lambda \in \mathbb{R}.$$

- (b) Such portfolios are thus efficient only if $\tilde{\mu}_t$, hence μ_t , is proportional to e .
- (c) If the initial value of the portfolio is $V_t = a_0 + \omega'_{\mathbf{a}} e = v$, its value at time $t+1$ is

$$V_{t+1} = a_0(1+r_t) + \omega'_{\mathbf{a}} y_{t+1} = v(1+r_t) + \omega'_{\mathbf{a}} (y_{t+1} - (1+r_t)e).$$

The conditional mean and variance of the solutions are thus given by

$$\begin{aligned} m_t = E_t V_{t+1} &= v(1+r_t) + \omega'_{\mathbf{a}} \tilde{\mu}_t = v(1+r_t) + \lambda e' \Omega_t^{-1} \tilde{\mu}_t, \\ \sigma_t^2 = \text{Var}_t(V_{t+1}) &= \omega'_{\mathbf{a}} \Omega_t \omega_{\mathbf{a}} = \lambda^2 e' \Omega_t^{-1} e. \end{aligned}$$

By eliminating λ we get

$$\sigma_t^2 = \left(\frac{m_t - (1+r_t)v}{e' \Omega_t^{-1} \tilde{\mu}_t} \right)^2 e' \Omega_t^{-1} e.$$

In the mean-variance graph, we obtain a semi-parabola located below the efficient frontier (with intersection on the vertical axis).

Exercise 2 :

1. See the course. The model is static, as the factor loading matrix is fixed.
2. We have $E(\mathbf{y}_t) = \mathbf{c}$, $V(\mathbf{y}_t) = \mathbf{B}\mathbf{\Gamma}\mathbf{B}' + \mathbf{\Sigma}$. For any vector $\mathbf{x} \neq 0$,

$$\mathbf{x}'V(\mathbf{y}_t)\mathbf{x} = \mathbf{x}'\mathbf{B}\mathbf{\Gamma}\mathbf{B}'\mathbf{x} + \mathbf{x}'\mathbf{\Sigma}\mathbf{x} > 0$$

because the latter term is strictly positive, while $\mathbf{x}'\mathbf{B}\mathbf{\Gamma}\mathbf{B}'\mathbf{x} = (\mathbf{B}'\mathbf{x})'\mathbf{\Gamma}(\mathbf{B}'\mathbf{x}) \geq 0$.

3. (a) We have, noting that $\mathbf{B}\mathbf{f}_t = \mathbf{I}_N\mathbf{B}\mathbf{f}_t$ where $\mathbf{I}_N$ is the identity matrix,

$$\mathbf{y}_t = \text{vec}(\mathbf{y}_t) = \text{vec}(\mathbf{B}\mathbf{f}_t) + \mathbf{u}_t = \underbrace{(\mathbf{f}_t' \otimes \mathbf{I}_N)}_{N \times NK} \underbrace{\text{vec}(\mathbf{B})}_{NK \times 1} + \mathbf{u}_t.$$

Hence, letting $\boldsymbol{\beta} = \text{vec}(\mathbf{B})$, we have $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U}$ where

$$\mathbf{X} = \mathbf{F} \otimes \mathbf{I}_N, \quad \mathbf{F} = \begin{pmatrix} \mathbf{f}_1' \\ \vdots \\ \mathbf{f}_n' \end{pmatrix}, \quad V(\mathbf{U}) = \begin{pmatrix} \mathbf{\Sigma} & 0 & 0 \\ & \ddots & \vdots \\ 0 & 0 & \mathbf{\Sigma} \end{pmatrix} = \mathbf{I}_N \otimes \mathbf{\Sigma} = \mathbf{\Sigma}_U.$$

- (b) The GLS estimator of $\boldsymbol{\beta}$ is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{\Sigma}_U^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Sigma}_U^{-1}\mathbf{Y} = \{(\mathbf{F}'\mathbf{F})^{-1}\mathbf{F}' \otimes \mathbf{I}_N\} \text{vec}(\tilde{\mathbf{Y}}) = \text{vec}(\tilde{\mathbf{Y}}\mathbf{F}(\mathbf{F}'\mathbf{F})^{-1})$$

where $\tilde{\mathbf{Y}} = \underbrace{[\mathbf{y}_1 \ \dots \ \mathbf{y}_n]}_{N \times n}$.

The estimator is independent of $\mathbf{\Sigma}_U$, therefore it coincides with the LS estimator.

- (c) We have $\hat{\boldsymbol{\beta}} = \text{vec}(\hat{\mathbf{B}}) = \text{vec}(\tilde{\mathbf{Y}}\mathbf{F}(\mathbf{F}'\mathbf{F})^{-1})$ where the matrices inside the vec have the same dimension. Thus

$$\hat{\mathbf{B}} = \tilde{\mathbf{Y}}\mathbf{F}(\mathbf{F}'\mathbf{F})^{-1} = \left(\sum_{t=1}^n \mathbf{y}_t \mathbf{f}_t' \right) \left(\sum_{t=1}^n \mathbf{f}_t \mathbf{f}_t' \right)^{-1}.$$

Exercise 3 : The p -values of the coefficients α are greater than 5% except for two assets. Thus we cannot reject the hypothesis that $\alpha_i = 0$, for all i . A global Wald test of the hypothesis that all α_i 's are equal to zero leads to rejection : the Wald statistics asymptotically follows a $\chi^2(6)$ distribution under the null and the p -value is less than 1%. From this test, we can't validate the CAPM. The β_i 's can be compared to 1 : $\beta_i > 1$ indicates an asset which is aggressive w.r.t. risk, while a value less than 1 indicates an asset which is conservative. Rows 4 and 5 in the table allow to analyze the systemic and idiosyncratic parts of risk for each asset. The last row provides the Sharpe ratios, which measure the performance of each stock : Asset 3 is the most performant here. According to the CAPM theory, if the market portfolio was efficient, its performance would be superior to those of all assets : this does not seem to be the case here (although we don't have confidence intervals for the estimation of the Sharpe ratios). This is not in contradiction with the rejection of the CAPM for this economy.



3A/Master/MS - first session
2024-25 - Semester 1

FINANCIAL ECONOMETRICS

Closed book examination, no calculator - 2 hours

Notations reminder : Consider an economy with N risky assets and one risk-free asset. Let $p_{i,t}$ the price of asset i at time t ($i = 1, \dots, N$) and let $y_{i,t+1} = \frac{p_{i,t+1}}{p_{i,t}}$ and $\mathbf{y}_{t+1} = (y_{1,t+1}, \dots, y_{N,t+1})'$. The risk-free asset has price 1 at time t and return r_t between time t and $t+1$. The conditional moments of $\mathbf{y}_{t+1}$ (given the available information at time t) are : $E_t(\mathbf{y}_{t+1}) = \boldsymbol{\mu}_t$, $\text{Var}_t(\mathbf{y}_{t+1}) = \boldsymbol{\Omega}_t$ where $\boldsymbol{\mu}_t \in \mathbb{R}^N$ and $\boldsymbol{\Omega}_t$ is a positive-definite matrix (denoted $\boldsymbol{\Omega}$ when independent of t).

Let $\mathbf{z}_{t+1} = (z_{i,t+1})_{i=1,\dots,N}$ and $\tilde{\boldsymbol{\mu}}_t = (\tilde{\mu}_{i,t})_{i=1,\dots,N}$ the vectors

$$\mathbf{z}_{t+1} = \mathbf{y}_{t+1} - (1 + r_t)\mathbf{e}, \quad \tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t - (1 + r_t)\mathbf{e}, \quad \text{where } \mathbf{e} = (1, \dots, 1)'.$$

Denote by V_t the value of a portfolio composed, at time t , of a_0 units of the risk-free asset and a_i units of assets i , for $i = 1, \dots, N$: $V_t = v = a_0 + \boldsymbol{\omega}'_a \mathbf{e}$ where $\boldsymbol{\omega}_a = (a_1 p_{1,t}, \dots, a_N p_{N,t})'$ and $\mathbf{a} = (a_1, \dots, a_N)'$.

The exercises are independent. Answers must be justified, even briefly.

Exercise 1 (7 points) :

- In this question all answers should be given without formulas.
 - Explain in a few lines what the efficient frontier (EF) is and how it is derived. Give its representation in the mean-variance space.
 - How does the location of an efficient portfolio on the EF depend on the risk aversion of the investor? How does the EF depend on the initial value v of the portfolios? Can a portfolio be located below/above the EF?
 - How is the curvature of the EF related to the market performance?
- Explain why we have $V_{t+1} = v(1 + r_t) + \boldsymbol{\omega}'_a \mathbf{z}_{t+1}$ and deduce $E_t(V_{t+1})$ and $\text{Var}_t(V_{t+1})$. For an efficient portfolio, show that these conditional moments (denoted m_t and σ_t^2) are related by $\sigma_t^2 = S_t^{-2} \{m_t - v(1 + r_t)\}^2$, where S_t is the market performance.
- Given observations $\mathbf{y}_1, \dots, \mathbf{y}_n$, explain how the EF can be estimated. Derive for each point of the EF an asymptotic confidence interval at the 95% significance level. *Hint : The following result can be used without proof : $\sqrt{n}\{\hat{S}_{t,n} - S_t\} \xrightarrow{d} \mathcal{N}(0, 1 + \frac{1}{2}S_t^2)$, where $\hat{S}_{t,n}$ is the usual estimator of S_t .*

Exercise 2 (7 points) Let $z_{t+1}(\mathbf{a})$ the net excess return of the risky part of a portfolio with composition $\mathbf{a}$ in risky assets : $z_{t+1}(\mathbf{a}) = \frac{\mathbf{a}'\mathbf{p}_{t+1}}{\mathbf{a}'\mathbf{p}_t} - (1 + r_t) = \bar{\boldsymbol{\omega}}_{\mathbf{a}}'\mathbf{z}_{t+1}$ where $\mathbf{p}_t = (p_{1t}, \dots, p_{Nt})'$, $\bar{\boldsymbol{\omega}}_{\mathbf{a}} = \frac{\boldsymbol{\omega}_{\mathbf{a}}}{\boldsymbol{\omega}_{\mathbf{a}}'\mathbf{e}}$.

Let two portfolios with compositions $\mathbf{a}$ and $\mathbf{b}$ in risky assets.

1. Consider the theoretical regressions :

$$\begin{aligned} z_{t+1}(\mathbf{b}) &= \alpha_t + \beta_t z_{t+1}(\mathbf{a}) + \epsilon_{t+1}, \\ z_{t+1}(\mathbf{a}) &= \gamma_t + \delta_t z_{t+1}(\mathbf{b}) + \varepsilon_{t+1}, \end{aligned}$$

where $E_t(\epsilon_{t+1}) = E_t(\varepsilon_{t+1}) = 0$, $\text{Cov}_t(\epsilon_{t+1}, z_{t+1}(\mathbf{a})) = \text{Cov}_t(\varepsilon_{t+1}, z_{t+1}(\mathbf{b})) = 0$, and $\alpha_t, \beta_t, \gamma_t, \delta_t$ are real numbers.

- (a) Compute $\alpha_t, \beta_t, \gamma_t, \delta_t$ in function of $\bar{\boldsymbol{\omega}}_{\mathbf{a}}, \bar{\boldsymbol{\omega}}_{\mathbf{b}}, \tilde{\boldsymbol{\mu}}_t$ and $\boldsymbol{\Omega}_t$.
 - (b) If portfolio $\mathbf{a}$ is efficient but $\mathbf{b}$ is not, show that one of the coefficients α_t and γ_t is null but the other is nonzero.
 - (c) If portfolio $\mathbf{a}$ is efficient, can we characterize the efficiency of portfolio $\mathbf{b}$ through the coefficient β_t ?
2. We have observations $z_t(\mathbf{a}), z_t(\mathbf{b})$, for $t = 1, \dots, n$.
 - (a) What are the least-squares estimators $\hat{\alpha}_n$ and $\hat{\beta}_n$ of the coefficients α and β (supposed to be independent of t)?
 - (b) Let $\sigma^2 = \text{Var}(\epsilon_t), \tilde{\mu}(\mathbf{a}) = E\{z_t(\mathbf{a})\}$, and $\sigma^2(\mathbf{a}) = \text{Var}\{z_t(\mathbf{a})\}$. Assume that

$$\sqrt{n} \begin{pmatrix} \hat{\alpha}_n - \alpha \\ \hat{\beta}_n - \beta \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(0, \begin{pmatrix} \left(1 + \frac{\tilde{\mu}^2(\mathbf{a})}{\sigma^2(\mathbf{a})}\right) \sigma^2 & \frac{-\tilde{\mu}(\mathbf{a})}{\sigma^2(\mathbf{a})} \sigma^2 \\ \frac{-\tilde{\mu}(\mathbf{a})}{\sigma^2(\mathbf{a})} \sigma^2 & \frac{1}{\sigma^2(\mathbf{a})} \sigma^2 \end{pmatrix} \right).$$

Propose a test of efficiency of the portfolio $\mathbf{a}$.

Exercise 3 (6 points) : Consider the Fama-French model : for $i = 1, \dots, N$ and $t = 1, \dots, n$,

$$z_{it} = c_i + B_{i1}z_t^{(m)} + B_{i2}\text{SMB}_t + B_{i3}\text{HML}_t + u_{it}.$$

1. Interpret the signs of the coefficients B_{i2} and B_{i3} . Write the model in matrix form. How would you estimate the factor loadings?
2. The model being unsatisfactory on some data set, it is proposed to consider a dynamic version of the model.
 - (a) Write the general form of the model and the assumptions made on loadings.
 - (b) An econometrician proposes to estimate the model by rolling regression. Is this approach consistent?
 - (c) Suggest an alternative approach for modelling the dynamic factor loadings.

SOLUTION

(NB : Not all questions are corrected in details)

Exercise 1 :

1. (a) The EF is obtained by solving the mean-variance optimization problem, for portfolios endowed with the same initial wealth. Efficient portfolios maximize the return for a given level of risk. The resulting curve is a semi-parabola in the mean-variance space. See the course for the graphic representation.
- (b) A portfolio's investor with low risk aversion will be far away from the vertical axis. The EF is translated vertically if v is modified. Portfolios located below the EF are non efficient. Portfolios located above the EF don't exist.
- (c) The curvature of the EF depends on the market performance S_t (see the formula defining the EF) : if S_t is large, an increase of the risk results in a larger increase of the expected return than when S_t is small.
2. See the course.
3. In the mean-variance space, the EF is defined by the equation $x = S_t^{-2}\{y - v(1 + r_t)\}^2$. The estimated EF is then given by $x = S_{t,n}^{-2}\{y - v(1 + r_t)\}^2$. We have

$$\sqrt{n}\{\hat{S}_{t,n}^{-2} - S_t^{-2}\} = \hat{S}_{t,n}^{-2}S_t^{-2}(\hat{S}_{t,n} + S_t)\sqrt{n}\{\hat{S}_{t,n} - S_t\} \stackrel{d}{=} 2S_t^{-3}\sqrt{n}\{\hat{S}_{t,n} - S_t\}$$

because $\hat{S}_{t,n} \rightarrow S_t$ in probability. It follows that

$$\sqrt{n}\{\hat{S}_{t,n}^{-2} - S_t^{-2}\} \xrightarrow{d} \mathcal{N}\left(0, 4S_t^{-6}\left(1 + \frac{1}{2}S_t^2\right)\right).$$

Thus, with probability converging to 95%, S_t^{-2} belongs to the interval

$$[a_n, b_n] = \left[\hat{S}_{t,n}^{-2} \left(1 - \frac{1.96}{\sqrt{n}} \sqrt{4 + 2\hat{S}_{t,n}^{-2}}\right), \hat{S}_{t,n}^{-2} \left(1 + \frac{1.96}{\sqrt{n}} \sqrt{4 + 2\hat{S}_{t,n}^{-2}}\right) \right].$$

Confidence intervals for the points of the EF can be deduced as the set of points between the curves $x = a_n\{y - v(1 + r_t)\}^2$ and $x = b_n\{y - v(1 + r_t)\}^2$.

Exercise 2 :

1. (a) We have $E_t(z_{t+1}(\mathbf{a})) = \bar{\omega}'_{\mathbf{a}}\tilde{\mu}_t$, $\text{Cov}_t(z_{t+1}(\mathbf{a}), z_{t+1}(\mathbf{b})) = \bar{\omega}'_{\mathbf{a}}\Omega_t\bar{\omega}_{\mathbf{b}}$. Hence

$$\beta_t = \frac{\bar{\omega}'_{\mathbf{a}}\Omega_t\bar{\omega}_{\mathbf{b}}}{\bar{\omega}'_{\mathbf{a}}\Omega_t\bar{\omega}_{\mathbf{a}}}, \quad \delta_t = \frac{\bar{\omega}'_{\mathbf{a}}\Omega_t\bar{\omega}_{\mathbf{b}}}{\bar{\omega}'_{\mathbf{b}}\Omega_t\bar{\omega}_{\mathbf{b}}}, \quad \alpha_t = \bar{\omega}'_{\mathbf{b}}\tilde{\mu}_t - \beta_t\bar{\omega}'_{\mathbf{a}}\tilde{\mu}_t, \quad \gamma_t = \bar{\omega}'_{\mathbf{a}}\tilde{\mu}_t - \delta_t\bar{\omega}'_{\mathbf{b}}\tilde{\mu}_t.$$

(b) If portfolio $\mathbf{a}$ is efficient, we have $\bar{\omega}_{\mathbf{a}} = \frac{\boldsymbol{\Omega}_t^{-1} \tilde{\mu}_t}{\tilde{\mu}_t' \boldsymbol{\Omega}_t^{-1} \mathbf{e}}$. Hence

$$\beta_t = \frac{\tilde{\mu}_t' \boldsymbol{\Omega}_t^{-1} \mathbf{e}}{\tilde{\mu}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\mu}_t} \tilde{\mu}_t' \bar{\omega}_{\mathbf{b}}, \quad \delta_t = \frac{\tilde{\mu}_t' \bar{\omega}_{\mathbf{b}}}{(\bar{\omega}_{\mathbf{b}}' \boldsymbol{\Omega}_t \bar{\omega}_{\mathbf{b}})(\tilde{\mu}_t' \boldsymbol{\Omega}_t^{-1} \mathbf{e})}, \quad \alpha_t = \bar{\omega}_{\mathbf{b}}' \tilde{\mu}_t - \beta_t \frac{\tilde{\mu}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\mu}_t}{\tilde{\mu}_t' \boldsymbol{\Omega}_t^{-1} \mathbf{e}} = 0.$$

Moreover

$$\gamma_t = \frac{\tilde{\mu}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\mu}_t}{\tilde{\mu}_t' \boldsymbol{\Omega}_t^{-1} \mathbf{e}} - \frac{(\tilde{\mu}_t' \bar{\omega}_{\mathbf{b}})^2}{(\bar{\omega}_{\mathbf{b}}' \boldsymbol{\Omega}_t \bar{\omega}_{\mathbf{b}})(\tilde{\mu}_t' \boldsymbol{\Omega}_t^{-1} \mathbf{e})}.$$

Thus $\gamma_t = 0$ if and only if $(\tilde{\mu}_t' \bar{\omega}_{\mathbf{b}})^2 = (\bar{\omega}_{\mathbf{b}}' \boldsymbol{\Omega}_t \bar{\omega}_{\mathbf{b}})(\tilde{\mu}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\mu}_t)$, which by the Cauchy-Schwarz inequality entails $\boldsymbol{\Omega}_t^{1/2} \bar{\omega}_{\mathbf{b}} = K \boldsymbol{\Omega}_t^{-1/2} \tilde{\mu}_t$ for some constant K , that is $\bar{\omega}_{\mathbf{b}} = K \boldsymbol{\Omega}_t^{-1} \tilde{\mu}_t$. If portfolio $\mathbf{b}$ is not efficient, we thus have $\gamma_t \neq 0$.

(c) If portfolios $\mathbf{a}$ and $\mathbf{b}$ are efficient, we have $\beta_t = 1$. However, $\beta_t = 1$ doesn't entail that $\mathbf{b}$ is efficient.

2. (a) Let

$$\hat{\mu}_n(\mathbf{a}) = \frac{1}{n} \sum_{t=1}^n z_t(\mathbf{a}), \quad \hat{\mu}_n(\mathbf{b}) = \frac{1}{n} \sum_{t=1}^n z_t(\mathbf{b}), \quad \hat{\sigma}_n^2(\mathbf{a}) = \frac{1}{n} \sum_{t=1}^n (z_t(\mathbf{a}) - \hat{\mu}_n(\mathbf{a}))^2.$$

The LS estimators of α and β are

$$\hat{\alpha}_n = \hat{\mu}_n(\mathbf{b}) - \hat{\beta}_n \hat{\mu}_n(\mathbf{a}), \quad \hat{\beta}_n = \frac{\frac{1}{n} \sum_{t=1}^n (z_t(\mathbf{b}) - \hat{\mu}_n(\mathbf{b}))(z_t(\mathbf{a}) - \hat{\mu}_n(\mathbf{a}))}{\hat{\sigma}_n^2(\mathbf{a})}.$$

(b) The null hypothesis is $H_0 : \alpha = 0$. The Wald statistic for H_0 is

$$W_n = \left(\left(1 + \frac{\hat{\mu}_n(\mathbf{a})}{\hat{\sigma}_n(\mathbf{a})} \right) \hat{\sigma}_n^2 \right)^{-1} n \hat{\alpha}_n^2$$

where $\hat{\sigma}_n^2$ is the empirical variance of the residuals. Under H_0 , $W_n \xrightarrow{d} \chi_1^2$. The critical region, at the asymptotic level $\alpha_0 \in (0, 1)$, is thus

$$C = \{W_n > \chi_1^2(1 - \alpha)\}$$

where $\chi_1^2(1 - \alpha_0)$ the $(1 - \alpha_0)$ quantile of the χ_1^2 law.

Exercise 3 :

1. A positive (resp. negative) B_{i3} means that asset i behaves like a value (resp. growth) asset. The matrix form of the model is $\mathbf{z}_t = \mathbf{c} + \mathbf{B}\mathbf{f}_t + \mathbf{u}_t$ with $\mathbf{f}_t = (z_t^{(m)}, \text{SMB}_t, \text{HML}_t)'$. The factor loadings can be estimated by LS equation by equation, without efficiency loss w.r.t. the GLS applied to the complete set of equations.

2. (a) A dynamic model model is of the form $\mathbf{z}_t = \mathbf{c}_t + \mathbf{B}_t \mathbf{f}_t + \mathbf{u}_t$, where $\mathbf{c}_t$ and $\mathbf{B}_t$ belong to the information set I_{t-1} (i.e. are functions of the returns and factors up to time $t - 1$).
- (b) See the course.
- (c) Two alternative approaches for modelling the dynamic factor loading are (i) the Dynamic Conditional Betas and (ii) the approach based on the Choleski decomposition of the conditional variance matrix.